Notes: Ramondo and Rodríguez-Clare (JPE, 2013)
Trade, Multinational Production, and the Gains from Openness

Contents

1 One-sentence summary

Ramondo and Rodríguez-Clare quantify the welfare gains from openness when trade and multinational production operate jointly, showing that the interaction between exporting, foreign production, intrafirm trade, and bridge multinational production changes the measured value of globalization.

2 Big picture

2.1 Research question

How large are the welfare gains from openness when countries interact through both trade and multinational production (MP), rather than through trade or MP alone?

2.2 Main object: openness

Openness includes exports, imports, foreign-affiliate production, intrafirm trade, and bridge multinational production (BMP), where an affiliate in a third country serves another foreign market.

2.3 Core mechanism

Trade and MP can be substitutes because either exports or local foreign production can serve a foreign market. They can also be complements because MP uses imported home-country inputs and because BMP combines MP with exporting from the affiliate location.

2.4 Model framework

The paper extends Eaton and Kortum's Ricardian trade model by allowing technologies to be used abroad through MP, with trade costs, MP costs, tradable intermediates, nontradable final goods, and potentially correlated productivity draws across countries.

2.5 Main quantitative result

Gains from openness are larger than gains from trade or gains from MP alone. In the calibrated model, average gains from openness are roughly 15–22 percent, while gains from trade and MP separately are smaller.

2.6 Key comparison

Trade-only models can understate the gains from trade because trade facilitates MP through intrafirm input flows. MP-only models can overstate gains from MP because trade and MP are substitutes for serving foreign markets.

2.7 Policy relevance

Trade agreements often liberalize both trade and FDI. The paper shows that evaluating trade or MP in isolation can mismeasure the value of such agreements.

3 Incremental contribution of the paper

3.1 Unifies trade and MP in one quantitative GE model

Earlier work often computed gains from trade or gains from MP separately. This paper builds a multicountry Ricardian model in which both flows coexist and interact.

3.2 Introduces bridge MP into gains-from-openness accounting

The model allows a multinational to produce in country l and export to country n, so MP can itself generate trade flows. This makes foreign affiliates potential export platforms rather than only local-market producers.

3.3 Separates complementarity from substitutability

The paper clarifies that trade and MP are not simply substitutes. Intrafirm trade and BMP make them complements, while market-serving FDI makes them substitutes.

3.4 Uses bilateral trade and bilateral MP data jointly

The calibration targets both kinds of flows across a sample of OECD economies, rather than estimating trade costs alone. This gives the model discipline on the two central channels of openness.

3.5 Connects MP to technology diffusion

MP is treated as a way of using a country's technology abroad, so gains from MP are gains from international technology sharing through firms.

3.6 Revises welfare measurement

The paper shows why trade-only formulas can miss important welfare gains when trade facilitates multinational activity. Similarly, MP-only comparisons can overstate MP gains when MP substitutes for exporting.

3.7 Provides country-level counterfactuals

The New Zealand exercise shows how changing trade and MP costs can generate very large welfare gains for small remote economies.

Detailed interpretation: The incremental contribution is not simply adding MP as another flow. The novelty is measuring how trade and MP change each other's welfare value. This makes the paper a benchmark for thinking about openness as a bundle of cross-border goods flows and cross-border production flows.

4 Data and measurement

4.1 Country sample and flows

The quantitative analysis uses a sample of 19 OECD countries. The key bilateral variables are trade shares and multinational production shares. Trade measures goods shipped across countries. MP measures gross production by foreign affiliates. The model also uses information on foreign affiliates' trade to construct bridge MP shares, especially for the United States.

- **Bilateral trade shares:** the share of country n's spending on goods produced in country i and shipped to n.
- **Bilateral MP shares:** the share of country n's production carried out by affiliates from country i.
- **Outward BMP share:** the share of affiliate output produced in a host country and exported to a third market.
- **Inward BMP share:** the share of production in a country by foreign affiliates that is exported onward.

Detailed interpretation: The data distinction between trade, MP, and BMP is crucial because the model treats them as different ways of using technologies across borders. A country can sell abroad directly, produce abroad locally, or produce in one foreign country and export to another.

4.2 Calibrated cost variables

Trade and MP costs depend on observable bilateral characteristics:

- distance;
- common border;
- common language;
- constants and dispersion parameters;
- sectoral MP parameters.

The paper estimates separate cost structures for trade and MP, under calibrations with productivity-draw correlation $\rho = 0$ and $\rho = 0.5$.

Detailed interpretation: The same geography can affect trade and MP differently. For example, distance may raise both costs, but a common border or language may matter more for one channel than the other.

4.3 Technology and preference parameters

The model contains an elasticity of substitution, a trade elasticity, sectoral cost shares, an intrafirm trade parameter, and a productivity-draw correlation parameter. The central calibrated parameters include θ, η, μ, ρ, and the intrafirm trade parameter a.

Detailed interpretation: The parameter ρ is especially important for bridge MP. If country-level productivity draws are highly correlated across goods, a country good at producing one intermediate is likely good at producing another, weakening the motive to use affiliates as export platforms.

5 Model and theoretical specification

5.1 Closed-economy cost structure

The model has tradable intermediate goods and nontradable final goods. Unit costs in the closed economy take the form

$$c^f = Aw^\alpha (P^g)^{1-\alpha}, \qquad c^g = Bw^\beta (P^g)^{1-\beta},$$

where w is the wage and P^g is the price index of composite intermediates. Prices are unit costs divided by productivity:

$$p^f(u) = \frac{c^f}{z^f(u)}, \qquad p^g(v) = \frac{c^g}{z^g(v)}.$$

Detailed interpretation: Intermediates enter both final-good and intermediate-good production, so trade and MP that lower intermediate costs can have economy-wide effects.

5.2 Open-economy channels

A country can serve a foreign market through three channels:

1. **Trade:** produce at home and export to the foreign market, paying trade costs.
2. **MP:** use home technology to produce in the foreign market, paying MP costs.
3. **Bridge MP:** use home technology to produce in country l and export from l to country n, paying both MP and trade costs.

Detailed interpretation: The third channel is the key model expansion relative to simple horizontal FDI models. It allows foreign affiliates to act as export platforms.

5.3 Fréchet productivity and cost minimization

As in Eaton-Kortum, countries have random productivity draws. The country and mode that delivers the good at the lowest delivered cost supplies the good. Delivered cost depends on productivity, wage, trade costs, and MP costs.

$$\text{supplier/mode} = \arg\min\{\text{delivered unit cost across countries and modes}\}.$$

Detailed interpretation: This creates a gravity-like structure for bilateral trade and MP shares. More productive countries, lower trade costs, and lower MP costs generate larger market shares.

5.4 Gains from openness, trade, and MP

The paper compares real wages across counterfactual equilibria:

$$GO_n = \frac{w_n/P_n \text{ with trade and MP}}{w_n/P_n \text{ under autarky}},$$

$$GT_n = \frac{w_n/P_n \text{ with trade and MP}}{w_n/P_n \text{ with MP only}},$$

$$GMP_n = \frac{w_n/P_n \text{ with trade and MP}}{w_n/P_n \text{ with trade only}}.$$

Detailed interpretation: Gains from trade are not measured from full autarky. They are measured as the benefit of adding trade on top of MP. Similarly, gains from MP are the benefit of adding MP on top of trade. This is what lets the paper measure interaction effects.

5.5 Complementarity and substitutability

Trade and MP are substitutes when a firm chooses either exports or local foreign production to serve a market. They are complements when MP relies on imported home inputs or when affiliate production is exported to third markets.

$$\text{Substitution: exports} \leftrightarrow \text{MP serving same market}$$

$$\text{Complementarity: MP} \rightarrow \text{intrafirm input trade and bridge exports.}$$

Detailed interpretation: The sign and magnitude of welfare interactions depend on which force dominates. The calibrated model finds complementarity is strong for trade gains, while substitutability slightly dominates for MP gains.

6 Identification and empirical design

This is a structural quantitative paper rather than a reduced-form causal design. Identification comes from matching multiple flows that impose restrictions on trade costs, MP costs, and technology parameters.

6.1 What identifies trade and MP costs

- Bilateral trade shares identify relative trade costs and trade-related geography.
- Bilateral MP shares identify relative MP costs and MP-related geography.
- US outward and inward BMP shares discipline the role of bridge MP and ρ.
- Intrafirm trade information disciplines the parameter a, which governs foreign affiliates' reliance on home inputs.

Detailed interpretation: Identification is overidentified in the sense that the model must simultaneously match bilateral trade, bilateral MP, aggregate trade/MP shares, and BMP patterns. The model fits trade and MP shares well but has more difficulty matching BMP levels when $\rho = 0.5$.

6.2 Role of alternative calibrations

The paper reports results for $\rho = 0$ and $\rho = 0.5$. With $\rho = 0$, productivity draws are independent across countries and goods. With $\rho = 0.5$, draws are correlated, which affects BMP and the mapping from trade elasticity to the model's θ parameter.

Detailed interpretation: Presenting both calibrations shows whether the welfare results depend on how correlated technologies are across country-mode combinations. The main qualitative result–large gains from openness–is robust, but BMP fit differs.

6.3 Validation moments

The model is checked against:

- means and standard deviations of bilateral trade shares;
- means and standard deviations of bilateral MP shares;
- correlations between trade and MP flows;
- outward and inward BMP shares for the United States;
- country-level trade and MP shares of GDP.

Detailed interpretation: The model fits bilateral trade and MP well, but it underpredicts some BMP patterns, especially for the United States. The authors emphasize this limitation when interpreting gains from bridge MP.

6.4 Main limitations

- The model focuses only on technology sharing through MP and goods trade; it excludes broader knowledge spillovers.
- The calibration uses developed OECD economies, so results may differ for developing countries.
- The model abstracts from firm-level heterogeneity in the Helpman-Melitz-Yeaple sense.
- The mapping from observed affiliate production to MP requires sectoral and measurement assumptions.
- BMP is difficult to match quantitatively, especially under correlated productivity draws.

Detailed interpretation: The results should be read as gains from observable openness through trade and MP, not as the total gains from all international knowledge diffusion.

7 Tables and figures

Every adjacent pair of figures/tables is placed on the same row. There are no landscape pages. If a single visual were left unpaired, it would be set to width 0.6 `\linewidth`, but all main visuals here are paired.

7.1 Figure 1: Cost structure in the closed economy; Table 1: Calibrated parameters

Read: Figure 1 shows how labor and composite intermediates feed both final-good and intermediate-good production. Table 1 reports the calibrated trade and MP cost parameters for $\rho = 0.5$ and $\rho = 0$.

Economic message: The figure gives the production backbone of the model, while the table gives the calibrated frictions that determine whether countries serve foreign markets through trade, MP, or bridge MP.

Detailed interpretation: Distance raises both trade and MP costs, but trade and MP costs respond differently to borders and language. The calibration therefore lets geographic characteristics affect the two channels differently.

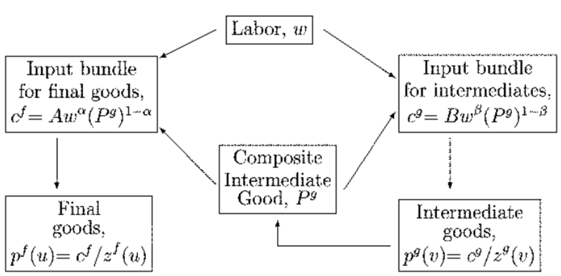


FIG. 1.—Cost structure in the closed economy

(a) Figure 1: cost structure in the closed economy.

TABLE 1
CALIBRATED PARAMETERS

	Model with $\rho = 0.5$		Model with $\rho = 0$	
	Trade	MP	Trade	MP
Cost parameters:				
Distance	.18	.26	.13	.31
Common border	.74	1.45	.65	1.86
Common language	.60	.40	.70	.36
Constant	.89	.95	1.09	1.30
Average costs	2.88	3.39	2.79	4.03
Standard deviation costs	.40	.47	.32	.48
Minimum cost	1.41	1.47	1.51	1.56
Maximum cost	5.49	7.12	4.69	8.45
Intrafirm trade parameter a	.15		.14	
MP cost parameter for final sector μ	1.55		1.11	
Variability parameter in Fréchet θ	3.75		4.40	

(b) Table 1: calibrated trade and MP cost parameters.

7.2 Table 2: Summary statistics: data and calibrated model; Table 3: Model's goodness of fit

Read: Table 2 compares model and data moments for bilateral trade shares, bilateral MP shares, correlations, and BMP shares. Table 3 reports R^2 and correlations for bilateral trade, MP, exports, imports, outward MP, and inward MP.

Economic message: These tables validate the quantitative model. The model matches bilateral trade and MP shares well, but it struggles more with BMP levels, especially the outward BMP share for the United States.

Detailed interpretation: The fit is strong for core bilateral moments: model R^2 values are about 0.89 for trade and about 0.64–0.66 for MP. This supports using the model for welfare counterfactuals, while the BMP mismatch is a warning about interpreting export-platform mechanisms too precisely.

Summary Statistics: Data and Calibrated Model

	Data	Model with $\rho = 0.5$	Model with $\rho = 0$
Bilateral trade shares:			
Average	.021	.021	.021
Standard deviation	.038	.035	.035
Bilateral MP shares:			
Average	.029	.024	.023
Standard deviation	.063	.038	.041
Correlation bilateral trade and MP shares	.701	.804	.793
Average outward BMP shares for the United States	.405	.115	.364
Average inward BMP shares for the United States	.090	.013	.065

(a) Table 2: data versus model summary moments.

TABLE 3
Model's Goodness of Fit

	Model with $\rho = 0.5$	Model with $\rho = 0$
Model's R^2:		
Bilateral trade shares	.89	.89
Bilateral MP shares	.64	.66
Correlations data and model:		
Bilateral trade shares	.93	.92
Bilateral MP shares	.76	.77
Total exports shares	.76	.71
Total imports shares	.85	.85
Total outward MP shares	.56	.54
Total inward MP shares	.56	.60

Note.—Bilateral MP is the gross value of production for affiliates from country i in l; total outward MP is the total gross value of production for foreign affiliates from country i; total inward MP is the total gross value of production for foreign affiliates in country l. Total exports and imports are expressed as shares of absorption in manufacturing.

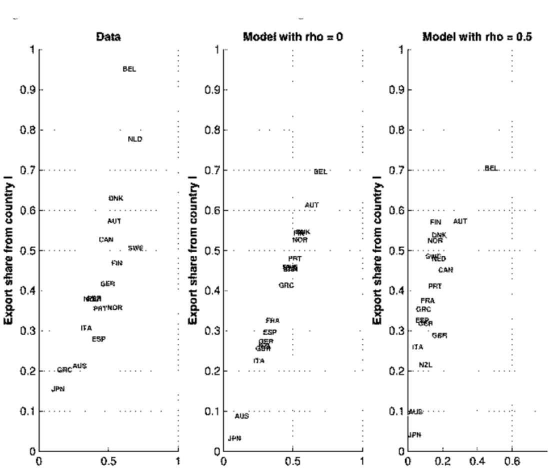
(b) Table 3: model goodness of fit.

7.3 Figure 2: BMP shares, data and calibrated model; Table 4: Gains from openness, trade, and MP under independence

Read: Figure 2 compares US outward BMP shares to export shares in the data and in the two model calibrations. Table 4 reports gains from openness, trade, and MP under the special independence case.

Economic message: Figure 2 shows why the BMP mechanism matters: export-platform activity is empirically related to exports. Table 4 shows that under independence, gains from openness can be much larger than trade gains alone.

Detailed interpretation: The data panel in Figure 2 has a strong positive relation between outward BMP and exports. The $\rho = 0$ calibration matches this pattern more closely than $\rho = 0.5$. Table 4 shows that small countries with high MP exposure, such as New Zealand and Portugal, have especially large gains from openness.



(a) Figure 2: BMP shares for the United States.

Gains from Openness, Trade, and Multinational Production: Independence

	GO^*_n	GT^*_n	GMP^*_n	L_n
New Zealand	1.365	1.053	1.296	.7
Denmark	1.137	1.096	1.037	1.2
Portugal	1.290	1.064	1.213	1.4
Canada	1.261	1.081	1.166	4.4
Germany	1.119	1.037	1.079	9.3
Japan	1.022	1.006	1.015	13.4
United States	1.053	1.015	1.037	28.7

Note.—The variables GO^*_n, GT^*_n, and GMP^*_n refer, respectively, to the gains from openness, trade, and multinational production, for country n, under independence. The variable L_n refers to the number of units of equipped labor in country n, as a percentage of OECD(19)'s total.

(b) Table 4: gains under independence.

7.4 Table 5: Gains from openness, trade, and MP: average; Table 6: Gains from openness, trade, and MP: selected countries

Read: Table 5 reports average gains across countries and sectors in the calibrated model. Table 6 reports country-level gains for New Zealand, Denmark, Portugal, Canada, Germany, Japan, and the United States.

Economic message: The average gains from openness are between 15 and 22 percent depending on calibration. Trade gains can exceed trade-only model gains because trade facilitates MP. MP gains can be lower than MP-only gains because trade and MP are substitutes as ways to serve foreign markets.

Detailed interpretation: Table 6 shows that country size and openness matter. Small or remote economies such as New Zealand and Portugal gain much more from openness than large economies such as the United States and Japan.

TABLE 5
GAINS FROM OPENNESS, TRADE, AND MULTINATIONAL PRODUCTION: AVERAGE

Average	GO_n	GT_n	GT^*_n	GMP_n	GMP^*_n
	Model with $\rho = 0$				
All sectors	1.148	1.080	1.062	1.086	1.091
Intermediate-goods sector	1.092	1.080	1.062	1.034	1.039
Final-goods sector	1.049	. . .	. . .	1.049	1.049
	Model with $\rho = 0.5$				
All sectors	1.221	1.101	1.074	1.095	1.116
Intermediate-goods sector	1.148	1.101	1.074	1.032	1.051
Final-goods sector	1.061	. . .	. . .	1.061	1.061

NOTE.—The variables GO_n, GT_n, and GMP_n refer, respectively, to the gains from openness, trade, and multinational production, for country n; GT^*_n and GMP^*_n refer to the gains from trade and multinational production, respectively, from trade-only and MP-only models.

(a) Table 5: average gains from openness, trade, and MP.

	GO_n	GT_n	GT^*_n	GMP_n	GMP^*_n	GMP^g_n
	Model with $\rho = 0$					
New Zealand	1.265	1.080	1.038	1.223	1.238	1.077
Denmark	1.199	1.119	1.097	1.102	1.109	1.042
Portugal	1.230	1.115	1.084	1.141	1.152	1.054
Canada	1.202	1.104	1.075	1.123	1.132	1.048
Germany	1.051	1.040	1.036	1.019	1.016	1.010
Japan	1.004	1.004	1.003	1.001	1.001	1.001
United States	1.012	1.010	1.008	1.005	1.004	1.003
	Model with $\rho = 0.5$					
New Zealand	1.320	1.100	1.037	1.221	1.262	1.096
Denmark	1.320	1.150	1.112	1.120	1.141	1.032
Portugal	1.384	1.140	1.082	1.192	1.219	1.063
Canada	1.282	1.127	1.090	1.128	1.157	1.046
Germany	1.079	1.056	1.051	1.018	1.026	1.005
Japan	1.007	1.005	1.005	1.001	1.002	1.000
United States	1.016	1.012	1.011	1.002	1.005	1.000

NOTE.—The variables GO_n, GT_n, GMP_n, and GMP^g_n refer, respectively, to the gains from openness, trade, multinational production, and multinational production in the intermediate-good sector for country n; GT^* and GMP^* refer to the gains from trade and multinational

(b) Table 6: selected-country gains from openness, trade, and MP.

7.5 Figure 3: Inward multinational production and complementarity; Table 7: New Zealand counterfactual

Read: Figure 3 plots the complementarity term $GT_n - GT_n^*$ against inward MP as a share of GDP. Table 7 asks how much New Zealand would gain if its trade and MP costs fell to Canadian or Belgian levels.

Economic message: Countries with high inward MP have higher gains from trade relative to trade-only models because trade supports MP through input flows and bridge production. The New Zealand counterfactual shows that lowering trade and MP costs jointly can generate very large real-wage gains.

Detailed interpretation: Figure 3 makes the paper's welfare mechanism visible: the gap between true gains from trade and trade-only gains rises with inward MP. Table 7 quantifies the policy relevance: New Zealand's real wage rises by 119 percent if both trade and MP costs fall to Canadian levels in the model.

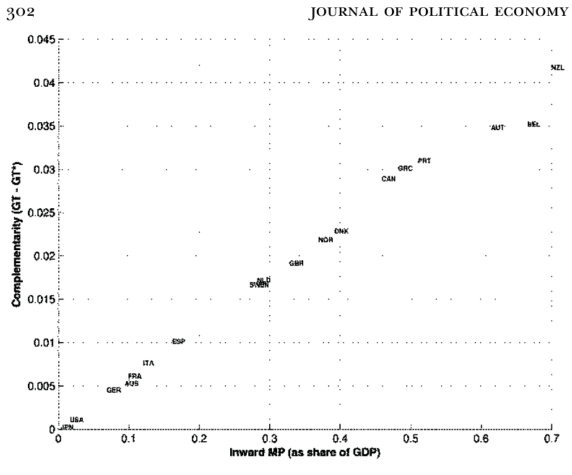


FIG. 3.—Inward multinational production and complementarity. The expression GT − GT* refers to the difference between the gains from trade calculated from the calibrated

(a) Figure 3: inward MP and complementarity in trade gains.

TABLE 7
GAINS FROM OPENNESS, TRADE, AND MULTINATIONAL PRODUCTION: NEW ZEALAND

	Percentage Change in Real Wage: New Zealand's Iceberg-Type Costs as in	
	Canada	Belgium
Trade	30	39
Multinational production	70	55
Trade and multinational production	119	75

NOTE.—Calculation using calibrated model with $\rho = 0$. Change in real wage for New Zealand of moving from the calibrated level of trade and MP costs to a situation in which (1) trade costs equal the ones calibrated for Canada (Belgium), (2) MP costs equal the ones calibrated for Canada (Belgium), and (3) both trade and MP costs equal the ones calibrated for Canada (Belgium).

(b) Table 7: New Zealand counterfactual gains.

8 Short summary

- The paper quantifies gains from openness when trade and multinational production coexist.
- It extends the Eaton-Kortum framework by allowing technologies to be used abroad.
- Trade and MP are substitutes when firms choose how to serve a foreign market.
- Trade and MP are complements through intrafirm trade and bridge MP.
- The model is calibrated to bilateral trade, bilateral MP, and BMP moments for OECD countries.
- Average gains from openness are much larger than gains from trade or MP alone.
- Trade-only models can understate gains from trade when trade facilitates MP.
- MP-only models can overstate gains from MP when trade and MP are substitutes.
- Small countries with high inward MP enjoy especially large gains from openness.
- The paper reframes openness as a combined system of goods flows and technology flows.

9 Conclusion

9.1 Core conclusion

Ramondo and Rodríguez-Clare show that countries gain from openness through the interaction of trade and multinational production. Measuring trade or MP in isolation can misstate welfare because each channel changes the value of the other. Trade facilitates MP by moving inputs and by enabling export-platform production. MP changes trade's value by offering an alternative way to serve foreign markets and by spreading technologies internationally.

9.2 Economic intuition

Trade and MP are jointly determined. If a firm can export or produce abroad, a fall in trade costs changes the incentive to use MP. If foreign affiliates rely on home-country inputs, trade makes MP more productive. If affiliates export to third markets, MP creates trade. The welfare effect of liberalization therefore depends on the whole network of trade and production links.

9.3 Incremental contribution revisited

The paper's contribution is to make the gains-from-openness calculation multidimensional. Previous trade-only models measure the cost of eliminating trade while holding MP absent. MP-only models do the reverse. This paper asks what happens when both channels are active and mutually affect each other. That is a more realistic benchmark for evaluating modern globalization.

9.4 Policy implication

Policies that reduce trade barriers and policies that reduce MP barriers should not be evaluated separately. Their interaction can be large. The New Zealand counterfactual shows that lowering both trade and MP costs can generate much larger gains than changing either one alone.

9.5 Limitations

The model does not capture all forms of international technology diffusion, such as licensing, spillovers, learning, or managerial transfers that do not appear as MP. It also underpredicts some BMP moments, especially for large economies. The estimates therefore measure gains from observable trade and MP, not the total gains from every channel of openness.

9.6 Takeaway

The main takeaway is that openness is a system. Trade, multinational production, imported inputs, and export-platform activity jointly determine welfare. Ignoring these interactions can lead to incorrect conclusions about the gains from globalization.